Grades 7 and 8 Physical Science

Properties, Changes, Bonding, and Conservation

Student Worksheets

Lessons 1 to 6 | October 14 to 22, 2026

Virginia Science Standards of Learning (2018): PS.3a

Name _____ Class _____

HOW EACH LESSON WORKS

1. Aim: the question you will answer by the end of class. **2. Do Now:** start right away, on your own. **3. Board Notes:** copy them into your science notebook. **4. Main Reading with Guided Questions:** read carefully and answer with your teacher. **5. Independent Practice:** one problem on your own. **6. Exit Check:** one quick answer before you leave. **Notebook / Extra Practice:** once done, work on this in your science notebook.

Bring your science notebook to every class.

STUDY FOR THE UNIT 4 TEST

Sentences marked like **ON THE TEST:** this (bold label, highlighted, and underlined) are on the Unit 4 Test, and so are the board notes you star in your notebook. Reread them, copy them into your science notebook, and study them before the test. They cover pure substances and mixtures (Lesson 1), intensive properties (Lesson 2), what density means (Lesson 3), and how to calculate density and identify a metal (Lesson 4). Lessons 5 and 6 are not on the Unit 4 Test; they prepare you for the density lab later in the unit.

LESSON 1 | WEDNESDAY, OCTOBER 14, 2026

Pure Substances and Mixtures

AIM: How can we classify matter as an element, a compound, or a mixture, and how can mixtures be separated?

WORDS TO KNOW	
pure substance	matter made of only one kind of particle, with the same makeup in every sample
element	a pure substance made of only one kind of atom
compound	a pure substance made of two or more elements chemically joined in a fixed ratio
mixture	two or more substances physically combined, in amounts that can vary
heterogeneous / homogeneous	a mixture that is not the same throughout / is the same throughout (a solution)

DO NOW | 3 minutes | Work on your own.

From Unit 2: What is an atom? Then name one element and one compound that you know.

MINI LESSON: BOARD NOTES | Copy all the board notes into your science notebook. **Put a star next to these notes in your notebook.**

ON THE TEST: **1.** Pure substance: one kind of particle **2.** Compound: elements joined in a fixed ratio (H_2O, sugar) **3.** Mixtures: salt water, air, trail mix

MAIN READING: ## What Is This Stuff Made Of?

Pure substances

Scientists sort matter into two big groups: pure substances and mixtures. **ON THE TEST:** A **pure substance** is made of only one kind of particle, and every sample of it has the same makeup. To decide, ask: is every particle the same? If yes, the material is a pure substance; if the particles are different, it is a mixture. Pure substances are either elements or compounds.

An **element** is made of only one kind of atom. Gold, oxygen, and aluminum are elements; each has its own box on the periodic table. Some elements, such as oxygen gas (O_2) and nitrogen gas (N_2), are made of molecules of two identical atoms; they are still elements because they have only one kind of atom. **ON THE TEST:** A **compound** is made of two or more elements chemically joined in a fixed ratio. Water and table sugar are compounds, so they are pure substances. Table sugar is a good example: every sugar particle is the same molecule, so sugar from any bag has the same makeup. Every water molecule has two hydrogen atoms joined to one oxygen atom, so the formula of water is always H_2O. A compound also has new properties: sodium (a soft, shiny metal) and chlorine (a poisonous green gas) join to form table salt, a white solid you can eat. Salt cannot be separated by picking, filtering, or evaporating; only a chemical change can do that.

Mixtures

A **mixture** is two or more substances that are physically combined but not chemically joined. Each substance keeps its own properties, and a mixture has no fixed recipe: one spoonful or five spoonfuls of sugar stirred into water are both sugar water.

Because the parts are not chemically joined, a mixture can be separated by physical means. When a solid is dissolved in a liquid, such as salt in water, evaporation removes the liquid as a gas and leaves the solid as a **residue** you can see. Not every mixture leaves a residue: liquids that evaporate easily are **volatile**, and a mixture of volatile liquids, such as water and pure alcohol, can dry up and leave nothing. So a cup that dries without residue might have held pure water or a mixture of liquids.

Two kinds of mixtures

In a **heterogeneous mixture**, a sample from one part is different from a sample from another part. In trail mix, sand in water, and oil floating on vinegar, you can see the different parts. Muddy water may look evenly brown, but if it sits for a while, the mud settles to the bottom.

In a **homogeneous mixture**, the parts are spread so evenly that the makeup is the same everywhere in the sample. **ON THE TEST:** Salt water, air, and clear apple juice are homogeneous mixtures, also called **solutions**. They are mixtures, not pure substances. Looking the same throughout does not make a material pure: salt water is still salt plus water, and the amount of salt can change.

Separating mixtures

Scientists choose a method by looking at how the parts differ. Pieces big enough to see, such as raisins in trail mix, can be **picked out by hand**. A magnet pulls iron filings out of sand, because only iron is attracted to it. A **filter**, such as a coffee filter, lets water through but traps sand. Filtering does not work for salt water, because the dissolved salt passes through the filter with the water; to get the salt back, let the water **evaporate**.

READING AND GUIDED QUESTIONS | Required | Answer during the reading.

1. **[VOCABULARY]** What is the difference between an element and a compound?

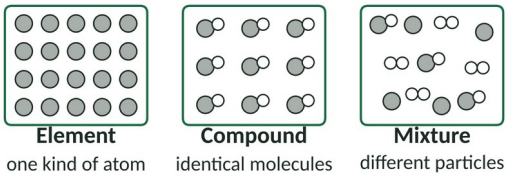


Figure 1. Particles of an element and a compound (both pure substances) and a mixture. Each circle is one atom; gray and white are two different elements. The element box shows one possible arrangement; some elements, such as O_2, are made of molecules.

2. **[INTERPRET THE FIGURE]** Look at Figure 1 above. Which box shows a mixture? Describe how its particles are different from the particles in the compound box.

INDEPENDENT PRACTICE | Required | 8 minutes | Work on your own.

3. **[APPLY]** Classify each material. Read the hint, then circle one answer in each column.

HOW TO DECIDE

1. Every particle the same? **No:** Mixture. **Yes:** go to step 2. **2.** Only one kind of atom in each particle? **Yes:** Element. **No:** Compound.

3. Mixtures only: the same throughout? **Homo**geneous. Not the same throughout? **Hetero**geneous.

Material	Hint	Circle one			Mixtures only	
Gold (Au)	*One symbol; only gold atoms.*	Element	Compound	Mixture	Homo	Hetero
Water (H_2O)	*Same H_2O molecule; two kinds of atoms.*	Element	Compound	Mixture	Homo	Hetero
Salt water	*Salt dissolved; amount can vary.*	Element	Compound	Mixture	Homo	Hetero
Trail mix	*Pieces you can see and pick out.*	Element	Compound	Mixture	Homo	Hetero
Carbon dioxide (CO_2)	*Same CO_2 molecule; two kinds of atoms.*	Element	Compound	Mixture	Homo	Hetero
Air	*N_2, O_2, and Ar gases spread evenly.*	Element	Compound	Mixture	Homo	Hetero

EXIT CHECK | 2 minutes: Name one pure substance and one mixture from today's reading.

NOTEBOOK / EXTRA PRACTICE | Once done, work on this. Complete it in your science notebook.

A. Why can a mixture be separated by physical means, such as filtering or evaporating, while a compound cannot? *(Write 3 to 4 sentences.)*

B. Think and Write: Omar leaves two cups of clear liquid open for a week. Cup A leaves white crystals behind; Cup B leaves nothing. What does each result tell you, and why might Cup B not have held pure water? *(Write 4 to 5 sentences. Draw both cups after drying and label what is left in each.)*

LESSON 2 | THURSDAY, OCTOBER 15, 2026

Physical Properties: Intensive and Extensive

AIM: How are intensive and extensive properties different, and why do intensive properties identify a substance?

WORDS TO KNOW	
physical property	a characteristic you can observe or measure without changing the substance
intensive property	a property that does not depend on the amount of the substance, such as color or density
extensive property	a property that depends on the amount of the substance, such as mass or volume

DO NOW | 3 minutes | Work on your own.

Classify each material from Lesson 1 as an element, a compound, or a mixture.

Oxygen gas (O_2): _____ Salt water: _____ Carbon dioxide (CO_2): _____

Explain one answer:

_____

MINI LESSON: BOARD NOTES | Copy all the board notes into your science notebook. **Put a star next to these notes in your notebook.**

ON THE TEST: **1.** Intensive: same for any sample size (density, color) **2.** Extensive: changes with amount (mass, volume, length)

MAIN READING: ## Properties You Can Observe and Measure

What is a physical property?

A **physical property** is a characteristic of a substance that you can observe or measure without changing the substance into something new. Shiny, orange-brown, and bendable are physical properties of a copper wire; looking at or bending the wire does not change it, and it is still copper.

Some physical properties can be observed with your senses: color, **luster** (how shiny a surface is), texture, and state of matter. Others need a tool or a test, such as mass, volume, density, hardness, melting point, and boiling point. **Hardness** describes how easily a material is scratched. A diamond can scratch glass, but glass cannot scratch a diamond.

Intensive and extensive properties

Some properties do not depend on how much of the substance you have. These are called **intensive properties**. A tiny drop of pure water and a full bathtub of pure water both freeze at 0 °C and boil at 100 °C at sea level. A small chip of copper and a long copper pipe have the same color and the same density, 8.96 g/cm^3. **ON THE TEST:** Color, luster, hardness, density, melting point, and boiling point are all intensive properties: they do not depend on the size of the sample. Density is the one to remember: a small bolt and a large bolt made of the same type of steel have about the same density under the same conditions, even though the large bolt has far more mass.

These temperatures come with two warnings. First, a boiling point depends on air pressure: in Denver, Colorado, where the air pressure is lower, water boils at about 94 °C instead of 100 °C. Second, impurities change both: at sea level pressure, salt water freezes below 0 °C and boils above 100 °C. Reference tables list values for pure substances at a stated temperature and pressure.

Other properties change when the amount changes. These are called **extensive properties**. **ON THE TEST:** Mass and volume are extensive. Length also changes with sample size, so it is treated as extensive: a short copper chip and a long copper pipe are the same material with very different lengths.

The half test

A simple test sorts any property. Imagine cutting the sample in half. If the property is different for each half, it is **extensive**; if it stays the same, it is **intensive**. Half a bottle of water has half the mass and half the volume, so mass and volume are extensive. Each half keeps the same color, density, and boiling point, so those properties are intensive.

Why intensive properties identify substances

Imagine two pieces of shiny metal, one large and one small. Comparing their masses will not tell you whether they are the same metal, because the large piece will have more mass either way. Intensive properties describe the substance itself. If both pieces have a density of 2.70 g/cm^3 and melt at 660 °C, they are very likely both aluminum. Scientists compare measured intensive properties with reference tables to identify unknown pure substances.

One property is usually not enough, because two substances can share a value. Salt and sugar are both white solids; water and pure isopropyl alcohol are both clear liquids. At room temperature and sea level pressure, water has a density of about 1.00 g/mL and boils at 100 °C, while pure isopropyl alcohol has a reference density of about 0.79 g/mL and boils at about 82 °C, so either property tells them apart. Store-bought rubbing alcohol is a mixture of isopropyl alcohol and water, so its values differ from both. Scientists check several intensive properties together; a match makes an identification very likely, not completely certain.

READING AND GUIDED QUESTIONS | Required | Answer during the reading.

1. **[COMPARE]** What is the difference between an intensive property and an extensive property?

2. **[APPLY]** Sort each property. Read the hint, then circle Intensive or Extensive.

HOW TO DECIDE

Use the half test: imagine cutting the sample in half. **Changes:** circle **Extensive**. **Stays the same:** circle **Intensive**.

Property	Hint	Circle one
Mass	*Does half a bottle of water have the same mass as a full one?*	Intensive Extensive
Color	*Is a small copper chip a different color from a long copper pipe?*	Intensive Extensive
Boiling point	*Does a drop boil at a different temperature than a bathtub?*	Intensive Extensive
Volume	*Does half a bottle of water take up the same space as a full one?*	Intensive Extensive
Density	*The copper chip and the copper pipe are both 8.96 g/cm³.*	Intensive Extensive
Length	*Compare a short copper chip and a long copper pipe.*	Intensive Extensive
Hardness	*Is a small diamond any easier to scratch than a big one?*	Intensive Extensive
Melting point	*Do a small ice cube and a big ice cube both melt at 0 °C?*	Intensive Extensive

INDEPENDENT PRACTICE | Required | 8 minutes | Work on your own.

3. **[USE EVIDENCE]** Amina has a large and a small piece of shiny metal. She says, "The large piece has more mass, so it must be a different metal." Do you agree? Use density in your answer. Write one or two sentences for each part.

Claim:

Evidence:

Reasoning:

NOTEBOOK / EXTRA PRACTICE | Once done, work on this. Complete it in your science notebook.

A. What is a physical property? Give one example. *(2 sentences)*

B. Half a bottle of water is poured out. What happens to its mass, volume, density, and boiling point? *(One sentence each)*

C. Liquid A: 20.0 g, 20.0 mL, boils at 100 °C. Liquid B: 31.6 g, 40.0 mL, boils at 82 °C. Which could be water and which pure isopropyl alcohol? Why can you not be certain? *(Show both density calculations, then 2 to 3 sentences.)*

LESSON 3 | FRIDAY, OCTOBER 16, 2026

Understanding Density: Mass, Volume, and Density

AIM: What do mass, volume, and density mean, and why do different materials have different densities?

WORDS TO KNOW

mass	the amount of matter in an object, measured in grams (g)
volume	the amount of space an object takes up, measured in milliliters (mL) or cubic centimeters (cm^3)
density	the mass in each unit of volume; density = mass ÷ volume
characteristic property	an intensive property, such as density, that helps identify a substance because it does not depend on the amount

DO NOW | 3 minutes | Work on your own.

A brick and a sponge are the same size. Which one has more mass? Explain why in one or two sentences.

MINI LESSON: BOARD NOTES | Copy all the board notes into your science notebook. **Put a star next to these notes in your notebook.**

ON THE TEST: **1.** Density: mass in each unit of volume **2.** Same material, any size: same density

MAIN READING: How Tightly Is Matter Packed?

Same size, different mass

A foam cup and a ceramic mug of the same size take up about the same space, but the mug has much more mass. Scientists describe this difference with **ON THE TEST:** **density**, which tells how much mass is packed into each unit of volume. In other words, density compares mass with volume: the mug has more mass packed into the same space, so it is denser.

Figure 1 shows three cubes that each have a volume of exactly 1 cm^3. The cork cube has a mass of only 0.24 g, while the lead cube has 11.3 g. Because each cube is exactly 1 cm^3, its mass in grams is also its density in grams per cubic centimeter.

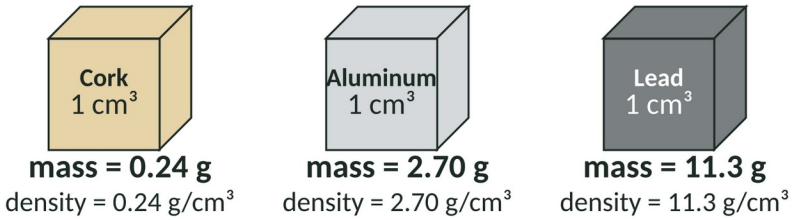


Figure 1. Three cubes with the same volume (1 cm^3) and very different masses.

Why different materials have different densities

Two things decide how much mass fits in a space: how much mass each particle has, and how closely the particles are packed. Lead atoms have much more mass than the particles in cork, and they are packed tightly together. Counting particles alone does not tell you the density, because particles of different substances can have very different masses.

Mass, volume, and density are different

Mass tells how much matter an object has, **volume** tells how much space it takes up, and **density** tells how much mass fits into each unit of that space. A 1 kg bag of feathers and a 1 kg rock have the same mass, but the bag of feathers, including the air spaces between the feathers, takes up far more space, so the bag as a whole is much less dense.

Volume has two common units. The volume of a solid is usually given in cubic centimeters (cm^3), and the volume of a liquid in milliliters (mL). Since 1 mL equals 1 cm^3, the two units can be compared directly. Density is written in grams per cubic centimeter (g/cm^3) or grams per milliliter (g/mL). "Per" means "in each."

Why density identifies substances

If you cut a uniform block of aluminum in half, each half has half the mass and half the volume, so the density stays the same, about 2.7 g/cm^3. **ON THE TEST:** Because density does not depend on the size of the sample, it is a **characteristic property.** A small piece and a large piece cut from the same aluminum have about the same density, 2.7 g/cm^3, under the same conditions: size changes the mass and the volume, but not the density. Scientists can compare a measured density with known values in a reference table to help identify an unknown pure substance.

READING AND GUIDED QUESTIONS | Required | Answer during the reading.

1. **[VOCABULARY]** In your own words, what does density tell you about a substance?

2. **[INTERPRET THE FIGURE]** Look at Figure 1 on page 6. All three cubes have the same volume. Which cube has the greatest density, and how do you know?

3. **[COMPARE]** A 1 kg bag of feathers and a 1 kg rock have the same mass. Which has the greater volume? Which has the greater density? Explain.

INDEPENDENT PRACTICE | Required | 8 minutes | Work on your own.

4. **[USE EVIDENCE]** Hana says, "A bigger object always has a bigger density." Do you agree? Use evidence from the reading.

Claim:

Evidence:

Reasoning:

EXIT CHECK | 2 minutes: Two blocks are the same size. Block A has more mass. Which block is denser?

NOTEBOOK / EXTRA PRACTICE | Once done, work on this. Complete it in your science notebook.

A. Which unit is used for the volume of a liquid, and how does it compare with a cubic centimeter? *(Write 1 to 2 sentences.)*

B. What is the difference between mass and volume? Give the unit for each. *(Write 2 to 3 sentences.)*

C. Think and Write: Explain why two objects of the same size can have different masses. Use the words particles, mass, volume, and density. *(Write 4 to 5 sentences. Draw two same-size boxes of particles and label the denser one.)*

LESSON 4 | DAY 1: MONDAY, OCTOBER 19 | DAY 2: TUESDAY, OCTOBER 20, 2026

Calculating Density and Identifying Substances

AIM: How do we calculate density step by step, and how can density help identify a substance?

WORDS TO KNOW

formula	a rule written with symbols; for density, D = m ÷ V
substitute	to replace each symbol in a formula with its number and unit
density units	g/cm^3 (grams per cubic centimeter) for solids or g/mL (grams per milliliter) for liquids
reference table	a list of known values, such as the densities of pure substances

MINI LESSON: BOARD NOTES | Copy all the board notes into your science notebook. **Put a star next to these notes in your notebook.**

ON THE TEST: **1.** Write D = m ÷ V (Day 1) **2.** Find the volume (block: length × width × height) (Day 2) **3.** Label g/cm^3; compare with a table to name the metal (Day 2)

DAY 1 | MONDAY, OCTOBER 19 | Calculating density with D = m ÷ V

DAY 1 DO NOW | 3 minutes | Work on your own.

A cork cube and a lead cube are both 1 cm^3. Which is denser? Explain using the words mass and volume.

_____

_____

MAIN READING: **Calculating Density Step by Step**

The six steps

ON THE TEST: To find density, divide the mass of a sample by its volume: **D = m ÷ V.** Mass always goes on top and volume on the bottom. A sample with a mass of 45 g and a volume of 15 cm^3 has a density of 45 g ÷ 15 cm^3 = 3.0 g/cm^3, so each cubic centimeter holds 3.0 g. Follow the same steps every time: **(1)** find the mass, **(2)** find the volume, **(3)** write the formula, **(4)** substitute the numbers with their units, **(5)** divide, and **(6)** write the answer with the correct density unit.

WORKED EXAMPLES

Solid: An aluminum block has a mass of 54 g and a volume of 20 cm^3. Mass = 54 g. Volume = 20 cm^3. D = m ÷ V. D = 54 g ÷ 20 cm^3. 54 ÷ 20 = 2.7. **D = 2.7 g/cm^3**. Every cubic centimeter holds 2.7 g of aluminum.

Liquid: Vegetable oil has a mass of 27.6 g and a volume of 30.0 mL. D = m ÷ V = 27.6 g ÷ 30.0 mL = **0.92 g/mL**.

READING AND GUIDED QUESTIONS | Required | Answer during the reading.

1. **[CALCULATE]** A piece of copper has a mass of 89.6 g and a volume of 10.0 cm^3. Calculate its density. (The first two steps are started for you.)

Formula: D = m ÷ V

Substitute (with units): D = 89.6 g ÷ _____

Calculate: _____

Answer (with unit): _____

INDEPENDENT PRACTICE | Required | 8 minutes | Work on your own.

2. **[CALCULATE]** A liquid sample has a mass of 46 g and a volume of 50 mL. What is its density?

Formula: _____

Substitute (with units): _____

Calculate: _____

Answer (with unit): _____

DAY 2 | TUESDAY, OCTOBER 20 | Finding volume and identifying a metal

DAY 2 DO NOW | 3 minutes | Work on your own.

A sample has a mass of 24 g and a volume of 8 cm^3. Find its density. Show the formula and the unit.

Finding the volume of a block

Sometimes the volume is not given. **ON THE TEST:** For a rectangular block, multiply its three measurements: **volume = length × width × height.** A block that is 2.0 cm long, 2.0 cm wide, and 2.0 cm tall has a volume of 2.0 × 2.0 × 2.0 = 8.0 cm^3. Multiply all three measurements, not just two, and label the answer in cubic centimeters (cm^3), because you multiplied three lengths in centimeters. Then use this volume in D = m ÷ V.

Using density to identify a substance

ON THE TEST: Because density is a characteristic property, a calculated density can be compared with a reference table to identify a metal. A shiny metal block with a mass of 81 g and a volume of 30 cm^3 has a density of 81 g ÷ 30 cm^3 = 2.7 g/cm^3. That matches aluminum (2.70 g/cm^3) far better than iron (7.87 g/cm^3) or copper (8.96 g/cm^3). To name the metal, find the table value closest to your answer. Density is strong evidence, but two materials can have similar densities, so it may not identify a substance by itself.

READING AND GUIDED QUESTIONS | Required | Answer during the reading.

3. **[CALCULATE]** A block is 5.0 cm long, 2.0 cm wide, and 1.0 cm tall. Find its volume.

Formula: _____

Substitute (with units): _____

Answer (with unit): _____

INDEPENDENT PRACTICE | Required | 8 minutes | Work on your own.

4. **[APPLY]** A metal block is 2.0 cm long, 2.0 cm wide, and 2.0 cm tall. Its mass is 63.2 g. Find its volume, then its density. Which metal could it be: aluminum (2.70 g/cm^3), iron (7.87 g/cm^3), or copper (8.96 g/cm^3)?

Volume (show l × w × h): _____

Formula: _____

Substitute (with units): _____

Calculate: _____

Answer (with unit): _____

Metal: _____

NOTEBOOK / EXTRA PRACTICE | Once done, work on this. Complete it in your science notebook.

A. Bilal cuts the aluminum block from the worked example (54 g, 20 cm^3) into four equal pieces. Find the mass, volume, and density of each piece. *(Show the formula, substitution, work, and answer with units.)*

B. A wooden block has a mass of 12.5 g and a volume of 25.0 cm^3. Calculate its density. *(Show the formula, substitution, work, and answer with units.)*

C. Think and Write: Explain why cutting a uniform aluminum block into smaller pieces changes its mass and volume but not its density. *(Write 3 to 4 sentences. Draw the block cut into four pieces and label one piece with its mass and volume.)*

LESSON 5 | WEDNESDAY, OCTOBER 21, 2026

Predicting Sinking and Floating

AIM: How can we predict whether an object will sink, float, or stay suspended in a liquid?

WORDS TO KNOW	
float	to rest at the surface because the object's average density is less than the liquid's
sink	to move to the bottom because the object's average density is greater than the liquid's
suspended	staying in place in a liquid, neither rising nor sinking, because its average density equals the liquid's
average density	an object's total mass divided by its total volume, including any air trapped inside

DO NOW | 3 minutes | Work on your own.

An object has a mass of 40 g and a volume of 8 cm^3. Calculate its density. Show the formula and units.

MINI LESSON: BOARD NOTES | Copy these notes into your science notebook. They are not on the Unit 4 Test.

KEY NOTES: **1.** Compare the object's average density with the liquid's density **2.** Less dense: floats. More dense: sinks **3.** Equal density: suspended (simplified model)

MAIN READING: Will It Sink or Float?

Comparing two densities

Density predicts whether an object sinks or floats in a liquid. To decide, compare two numbers: the object's density and the liquid's density. Water has a density of about 1.0 g/mL, so an object with a density below 1.0 g/cm^3 floats in water and an object above 1.0 g/cm^3 sinks. Cork (0.24 g/cm^3) floats, while a copper coin (8.96 g/cm^3) sinks.

What matters is the object's **average density**: its total mass divided by its total volume, including any air trapped inside. In these problems the liquid is still, and the object does not soak up the liquid or dissolve. An object **less dense** than the liquid floats. An object **more dense** than the liquid sinks. In this simplified model, an object with the **same** average density as the liquid stays **suspended**: it neither rises nor sinks. In real measurements, an object whose density is very close to the liquid's may drift slowly up or down.

Density, not weight

Sinking depends on density, not on how heavy an object is. A huge log floats on a lake while a tiny pebble sinks, because wood is less dense than water and rock is more dense. A steel ship floats because the air inside its hollow hull lowers its average density below that of water.

An unusual case: ice

Most substances are denser as solids than as liquids, but ice (about 0.92 g/cm^3) is less dense than liquid water (1.00 g/mL). That is why ice cubes float and ponds freeze from the top down.

Liquids in layers

Liquids can float on other liquids too. Liquids that do not mix, such as vegetable oil and water, can form layers, with the denser liquid at the bottom: vegetable oil (0.92 g/mL) floats on water (1.00 g/mL). Honey (1.42 g/mL) is denser than water, but honey mixes with water, so it does not form a lasting layer under it. An object dropped in sinks through any layer less dense than itself and rests on top of the first layer that is more dense.

Same object, different liquid

Whether an object floats depends on the liquid as well as the object. A fresh egg sinks in a glass of tap water, because the egg is slightly denser than water. Stir in plenty of salt and the salt water becomes denser than the egg, so the egg floats. The egg did not change; only the density of the liquid did.

READING AND GUIDED QUESTIONS | Required | Answer during the reading.

1. **[KEY DETAIL]** How can you predict whether an object will sink, float, or stay suspended in a liquid?

2. **[APPLY]** Water has a density of 1.0 g/mL. For each object, write <, =, or > in the hint, then circle Sink, Float, or Suspended. Assume still water and no object soaks up water.

HOW TO DECIDE

Compare the object with water (1.0). **Less than 1.0:** Float. **Greater than 1.0:** Sink. **Equal to 1.0:** Suspended.

Object	Density (g/cm³)	Hint: compare with water	Circle one
Cork	0.24	0.24 ___ 1.0	Sink Float Suspended
Aluminum block	2.70	2.70 ___ 1.0	Sink Float Suspended
Ice cube	0.92	0.92 ___ 1.0	Sink Float Suspended
Plastic button	1.20	1.20 ___ 1.0	Sink Float Suspended
Candle wax	0.90	0.90 ___ 1.0	Sink Float Suspended
Sealed plastic bead	1.00	1.00 ___ 1.0	Sink Float Suspended
Dry pine wood	0.50	0.50 ___ 1.0	Sink Float Suspended

INDEPENDENT PRACTICE | Required | 8 minutes | Work on your own.

3. **[APPLY]** A cup holds a layer of water (1.00 g/mL) with vegetable oil (0.92 g/mL) on top. Where will a plastic bead with a density of 0.95 g/cm³ come to rest? The liquids are still, and the bead does not soak up either liquid. Compare the bead with each liquid.

EXIT CHECK | 2 minutes: An object has a density of 1.3 g/cm³. Will it sink or float in still water? Why?

NOTEBOOK / EXTRA PRACTICE | Once done, work on this. Complete it in your science notebook.

A. Why does ice float on water? Why do scientists call this an unusual case? *(Write 2 to 3 sentences.)*

B. Think and Write: Explain why a large log floats while a small pebble sinks. Then explain how a steel ship can float even though steel is denser than water. *(Write 4 to 5 sentences. Draw the log, the pebble, and the ship in water and label each.)*

LESSON 6 | THURSDAY, OCTOBER 22, 2026

Water Displacement and Density Applications

AIM: How can water displacement help us find the volume and density of an irregular object and identify what it is made of?

WORDS TO KNOW

irregular object	an object with an uneven shape, such as a rock, that is hard to measure with a ruler
water displacement	finding an object's volume from how much water it pushes aside
displaced volume	final water level minus starting water level; equal to the object's volume
submerged	completely under the water

DO NOW | 3 minutes | Work on your own.

A cork (0.24 g/cm^3) and a penny (about 7 g/cm^3) are dropped into still water. Which floats, and which sinks? Explain.

MINI LESSON: BOARD NOTES | Copy these notes into your science notebook. They are not on the Unit 4 Test.

KEY NOTES: **1.** Displaced volume = final level − starting level **2.** 1 mL = 1 cm^3 **3.** Density = mass ÷ volume **4.** Object must be fully submerged and must not absorb water, dissolve, or react with it

MAIN READING: ## Letting Water Measure the Volume

Why a ruler is not enough

A ruler works for a box: you measure the length, width, and height and multiply. A rock, a bolt, or a piece of jewelry has an uneven shape, so there is no simple length, width, and height to measure. For an **irregular object**, scientists let water do the measuring instead.

How water displacement works

When an object is completely under water, it pushes aside, or **displaces**, a volume of water equal to its own volume. To use this: read the starting water level, gently lower the object until it is completely **submerged**, and read the final level. The **displaced volume** is the final level minus the starting level. Because 1 mL = 1 cm^3, a rise of 12 mL means the object has a volume of 12 cm^3.

The method works only if the object is completely under water and does not soak up water, dissolve, or react with the water. A floating object is only partly under water, so it does not displace its full volume.

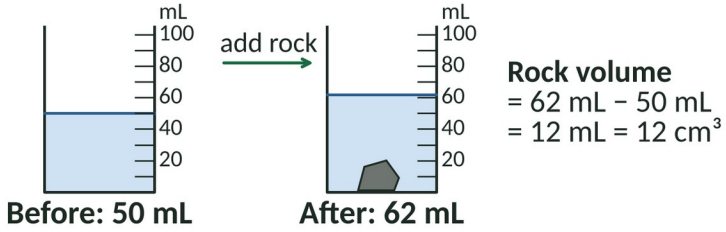


Figure 1. The water level rises from 50 mL to 62 mL when the rock is added.

WORKED EXAMPLE

Starting level = 50 mL. Final level = 62 mL. Volume = 62 mL − 50 mL = 12 mL = 12 cm^3.

The rock has a mass of 31.2 g. D = m ÷ V = 31.2 g ÷ 12 cm^3 = **2.6 g/cm^3**.

Using displacement to identify a material

Once you know the mass and the displaced volume, you can calculate the density and compare it with a reference table. A density that matches a known metal is strong evidence for what the object is made of, and checking one more property, such as color, makes the evidence stronger.

READING AND GUIDED QUESTIONS | Required | Answer during the reading.

1. **[KEY DETAIL]** Why must an object be completely under water, and not soak up, dissolve in, or react with the water, when you measure its volume by water displacement?

2. **[CALCULATE]** Omar's measuring cup holds 40 mL of water. He drops in a small piece of metal, and it sinks completely. The water rises to 47 mL. The metal has a mass of 62.7 g. Find the volume and density, then identify the metal. (Aluminum 2.70 g/cm^3, iron 7.87 g/cm^3, copper 8.96 g/cm^3, gold 19.3 g/cm^3)

Volume (final minus start): _____

Formula: _____

Substitute (with units): _____

Calculate: _____

Answer (with unit): _____

Metal: _____

INDEPENDENT PRACTICE | Required | 8 minutes | Work on your own.

3. **[CALCULATE]** A stone sinks completely and raises the water from 30 mL to 35 mL. Its mass is 13.0 g. Find its volume and density.

Volume (final minus start): _____

Formula: _____

Substitute (with units): _____

Calculate: _____

Answer (with unit): _____

EXIT CHECK | 2 minutes: The water rises from 20 mL to 26 mL when an object is added. What is the object's volume?

_____

NOTEBOOK / EXTRA PRACTICE | Once done, work on this. Complete it in your science notebook.

A. A metal bolt (31.5 g) sinks completely and raises the water from 100.0 mL to 104.0 mL. Find its density, then use the metals in question 2 to identify it. *(Show the formula, substitution, work, and answer with units.)*

B. Think and Write: Explain, step by step, how water displacement helps identify an unknown metal. Use question 2 as evidence. *(4 to 5 sentences)*